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(*======================================================================
Readings:

    This lab has no prerequisite textbook reading.

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(*
                             CS51 Lab 20
                              Synthesis
 *)

type image = float list list ;;
(* images are lists of lists of floats between 0. (white) and 1. (black) *)
type size = int * int ;;
open Graphics ;;

(* threshold thershold image -- image where pixels above the threshold
value are black *)
let threshold img threshold =
  List.map  (fun row -> List.map (fun v -> if v <= threshold then 0. else 1.)
                                 row) img

(* show the image *)
let depict img =
  Graphics.open_graph ""; Graphics.clear_graph ();
  let x, y = List.length (List.hd img), List.length img in Graphics.resize_window x y;
  let depict_pix v r c = let lvl = int_of_float (255. *. (1. -. v)) in Graphics.set_color
  plot c (y - r) in
  List.iteri (fun r row -> List.iteri (fun c pix -> depict_pix pix r c) row) img;
  Unix.sleep 2; Graphics.close_graph () ;;

(* dither max image -- dithered image *)
let dither img =
  List.map
    (fun row ->
     List.map
       (fun v -> if v > Random.float 1.
                 then 1.
                 else 0.) row)
    img

let mona = Monalisa.image ;;

  depict mona ;;

  let mona_threshold = threshold mona 0.75 ;;
    depict mona_threshold ;;

    let mona_dither = dither mona ;;
      depict mona_dither ;;
```